

# Shubham Bhattacharya

Backend Engineer (Node.js / TypeScript)

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[github.com/shubhambhattacharya-dev](https://github.com/shubhambhattacharya-dev) | [linkedin.com/in/shubhambhattadev](https://linkedin.com/in/shubhambhattadev) | [shubhambhattacharya.dev](https://shubhambhattacharya.dev)

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## SKILLS

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|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>                | TypeScript, JavaScript (Node.js), SQL                                                                          |
| <b>Backend &amp; API:</b>        | Express.js, Next.js, REST API Design, Middleware, JWT Auth, Rate Limiting, WebSockets                          |
| <b>Databases:</b>                | PostgreSQL, Prisma ORM, Redis                                                                                  |
| <b>AI &amp; LLM Integration:</b> | Custom ReAct Loops, Multi-Model Routing, Zod Validation, Prompt Injection Protection, tiktoken Cost Estimation |
| <b>DevOps &amp; Tools:</b>       | GitHub Actions, Docker Compose, Vercel, Render                                                                 |
| <b>Observability:</b>            | Sentry Error Tracking, Pino Structured Logging                                                                 |

## PROJECTS

### AI Ops — Enterprise AI Tools (Node.js, TypeScript, Express, Groq SDK, PostgreSQL, Prisma, Zod, Sentry)

[github.com/shubhambhattacharya-dev/Agentic\\_Project](https://github.com/shubhambhattacharya-dev/Agentic_Project) | [Live Demo](#)

- Built a **custom ReAct orchestration engine** for autonomous order-tracking across multi-step reasoning cycles without relying on LangChain.
- Designed a **Zod-validated tool registry** enforcing strict schema validation on every LLM output prior to database operations.
- Implemented multi-model fallback routing (*Llama 3.3-70B* → *Llama 3.1-8B*) via Groq SDK to maintain uptime during provider rate limits.
- Integrated Sentry error tracking and Pino structured logging with **tiktoken-based cost accounting** for backend observability.

### Medo Copilot — Multi-Provider AI Audit Platform (Next.js, TypeScript, Vercel AI SDK, Clerk, Gemini, Groq, OpenRouter)

[github.com/shubhambhattacharya-dev/Medo\\_Copilot](https://github.com/shubhambhattacharya-dev/Medo_Copilot) | [medo-copilot.shubhambhattacharya.dev](https://medo-copilot.shubhambhattacharya.dev)

- Engineered a **3-tier LLM waterfall system** (*Gemini 2.0* → *Groq/Llama 4 Scout* → *OpenRouter/Claude*) with automated provider failover.
- Designed a hybrid scoring engine combining AI code analysis with Lighthouse performance metrics and static security scans.
- Developed automated Playwright screenshot capture for **multimodal LLM visual inspection**.
- Built a static security analyzer to detect hardcoded secrets, code vulnerabilities, and deployment risks.

### DocNow — Telemedicine SaaS Platform (Next.js, TypeScript, Clerk, Prisma, PostgreSQL, Vonage WebRTC)

[github.com/shubhambhattacharya-dev/health](https://github.com/shubhambhattacharya-dev/health) | [docnow-zeta.vercel.app](https://docnow-zeta.vercel.app)

- Built an intelligent symptom triage classifier routing patients to medical specialists based on reported symptoms.
- Developed browser-based WebRTC video consultation streams via Vonage API with zero client installation required.
- Architected a credit wallet system with an **atomic PostgreSQL audit trail** for purchases, deductions, and transaction history.
- Designed multi-role access control dashboards (*Patient, Doctor, Admin*) paired with an automated payout engine.

## EDUCATION

### Bachelor of Computer Applications (BCA)

Expected Dec 2026

Parul University, Vadodara, Gujarat

CGPA: 8.13 | SGPA: 8.45 | First Class with Distinction

**Relevant Coursework:** Soft Computing (AI), Software Engineering, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks.